Real Analysis II

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Class 1, January 9 2026

1 Riemann Integration

The physical meaning of an integral of a function is the area under the curve with the x-axis. We intend to compute the said area by dividing the region into rectangular chunks.
Let $I = [a, b]$ be an interval. A partition P of I is a finite subset of I that contains a, b. For any partition P there are $x_0, x_1, \ldots, x_n$ such that
$a = x_0 < \cdots < x_n = b$. We write $P = \{x_0, \ldots, x_n\}$. Let $f : I \to \mathbb{R}$ be a bounded function. Let m, M be such that $m \leq f(x) \leq M$ for all $x \in I$. Let $P = \{x_0, \ldots, x_n\}$ be a partition of I. Take $I_k = [x_{k-1}, x_k]$ and call:

- $\Delta x_k = x_{k+1} - x_k$
- $m_k = \inf_{x \in I_k} f(x)$
- $M_k = \sup_{x \in I_k} f(x)$

Define

$$\mathcal{L}(f, P) = \sum_{k=1}^{n} (\Delta x_k) m_k$$

$$\mathcal{U}(f, P) = \sum_{k=1}^{n} (\Delta x_k) M_k$$

Clearly, $\mathcal{L}(f, P) \leq \mathcal{U}(f, P)$. If P and Q are two partitions of I. We say that Q is a refinement of P if $P \subseteq Q$. We have the following result:

Lemma 1. *In the above situation,* $\mathcal{L}(f, P) \leq \mathcal{L}(f, Q) \leq \mathcal{U}(f, Q) \leq \mathcal{U}(f, P)$.

We will prove this later on.

Let

$$\underline{\int_a^b} f(x)\ dx = \sup_P \mathcal{L}(f, P)$$

be the Lower Riemann Sum and

$$\overline{\int_a^b} f(x)\ dx = \inf_P \mathcal{U}(f, P)$$

A bounded function f in $I = [a, b]$ is called Riemann Integrable if the Lower Riemann Sum and the Upper Riemann sum are the same. In this case we write $f \in \mathcal{R}[a, b]$.
Let P_1 and P_2 be two partitions of I. Let $P = P_1 \cup P_2$. P is a partition and its called a common refinement of P_1 and P_2.

Proof(Lemma 1). Let $P = \{x_0 < \cdots < x_n\}$ and let Q consist of just one more point y. Let i be such that $x_{i-1} < y < x_i$.
Let

$$w_1 = \inf f(x),\ x_{i-1} \leq x \leq y$$

and,

$$w_2 = \inf f(x),\ y \leq x \leq x_i$$

Clearly, we have $w_1 \geq m_i$ and $w_2 \geq m_i$, where $m_i = \inf f(x)$, $x \in [x_{i-1}, x_i]$
Then

$$\mathcal{L}(f, Q) - \mathcal{L}(f, P) = w_1(y - x_{i-1}) + w_2(x_i - y) - m_i(x_i - x_{i-1})$$

$$= (w_1 - m_i)(y - x_{i-1}) + (w_2 - m_i)(x_i - y) \geq 0$$

and hence the first part of the chain is proved. That $\mathcal{L}(f, Q) \leq \mathcal{U}(f, Q)$ is trivial. The last part can be proven analogously. □

Proposition 1.

$$\underline{\int_a^b} f(x)\ dx \leq \overline{\int_a^b} f(x)\ dx$$

Proof. Let P_1 and P_2 be two partitions with P^* being their common refinement. Then we have:

$$\mathcal{L}(f, P_1) \leq \mathcal{L}(f, P^*) \leq \mathcal{U}(f, P^*) \leq \mathcal{U}(f, P_2)$$

. Keeping P_2 fixed and taking supremum over all P_1 yields

$$\underline{\int_a^b} f(x)\ dx \leq \mathcal{U}(f, P_2)$$

finally taking infimum over all P_2 we get our desired result. □

Proposition 2. *$f \in \mathcal{R}[a,b]$ iff for each $\epsilon > 0$ there is a partition P such that*

$$|\mathcal{U}(f,P^*) - \mathcal{L}(f,P^*)| < \epsilon$$

for any refinement P^* of P.

Proof. Assume first that f is Riemann integrable over I. Let $\epsilon > 0$.

$\int_a^b f = \inf$

For the converse, let us assume that for each $\epsilon > 0$ there is a partition P such that

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) < \epsilon$$

then

$$\inf_P \mathcal{U}(f,P) - \sup_P \mathcal{L}(f,P) \leq \mathcal{U}(f,P) - \mathcal{L}(f,P) < \epsilon$$

and

$$0 \leq \overline{\int_a^b} f(x)\ dx - \underline{\int_a^b} f(x)\ dx < \epsilon$$

Let $\epsilon \to 0$. Then we get

$$\overline{\int_a^b} f(x)\ dx = \underline{\int_a^b} f(x)\ dx$$

and thus $f \in \mathcal{R}[a,b]$. □

This is analogous to the Cauchy Criterion for sequences.

Class 2, January 13 2026

1.1 Properties

Proposition 3. *If $f, g \in \mathcal{R}[a,b]$ then $\alpha f + \beta g \in \mathcal{R}[a,b]$, for any $\alpha, \beta \in \mathbb{R}$.*

Proof. $f \in \mathcal{R}[a,b] \implies \exists$ partitions P_1, P_2 such that

$$\mathcal{U}(f,P_1) - \mathcal{L}(f,P_1) < \epsilon/2$$

and,

$$\mathcal{U}(g,P_2) - \mathcal{L}(g,P_2) < \epsilon/2$$

Let $P = P_1 \cup P_2$. Then

$$\int_a^b f(x)\ dx - \mathcal{L}(f,P) \leq \mathcal{U}(f,P) - \mathcal{L}(f,P) < \epsilon/2$$

and similarly,

$$\int_a^b g(x)\ dx \leq \mathcal{U}(g,P) - \mathcal{L}(g,P) < \epsilon/2$$

We note that

$$\mathcal{L}(f+g,P) \geq \mathcal{L}(f,P) + \mathcal{L}(g,P)$$

and

$$\mathcal{U}(f+g,P) \leq \mathcal{U}(f,P) + \mathcal{U}(g,P)$$

And thus,

$$\int_a^b f(x)\ dx + \int_a^b g(x)\ dx - \mathcal{L}(f+g,P) \leq \int_a^b f(x)\ dx + \int_a^b g(x)\ dx - (\mathcal{L}(f,P) + \mathcal{L}(g,P)) < \epsilon$$

We get a similar bound for the upper sum. This finally yields

$$\mathcal{U}(f+g,P) - \mathcal{L}(f+g,P) < 2\epsilon$$

. Since ϵ was arbitrary, $f+g \in \mathcal{R}[a,b]$. Now since

$$\int_a^b f+g = \sup_Q \mathcal{L}(f+g,Q)$$

we have

$$\int_a^b f + \int_a^b g - \int_a^b (f+g) \leq \int_a^b f + \int_a^b g - \mathcal{L}(f+g,P) < \epsilon$$

letting $\epsilon \to 0$ we get

$$\int_a^b f + \int_a^b g - \int_a^b (f+g) \leq 0$$

and similarly we have the reverse inequality. With this, we conclude that

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g$$

As for the scalar multiple case, let $\alpha \in \mathbb{R}$, $f \in \mathcal{R}[a,b]$. Let $P = \{x_0, \ldots, x_n\}$ be a partition.

Case 1: $\alpha \geq 0$. Then

$$\mathcal{L}(\alpha f, P) = \sum \inf_{x \in I_k} \alpha f(x) \Delta x_k$$

$$= \sum \alpha \inf f(x) \Delta x_k = \alpha \mathcal{L}(f,P)$$

Similarly, $\mathcal{U}(\alpha f, P) = \alpha \mathcal{U}(f,P)$. Since this implies

$$\underline{\int_a^b} \alpha f = \alpha \underline{\int_a^b} f = \alpha \int_a^b f$$

and,

$$\overline{\int_a^b} \alpha f = \alpha \overline{\int_a^b} f = \alpha \int_a^b f$$

so

$$\int_a^b \alpha f = \alpha \int_a^b f$$

□

Proposition 4. *Let $f \in \mathcal{R}[a,b]$, $c \in [a,b]$. Then $f \in \mathcal{R}[a,c], f \in \mathcal{R}[c,b]$ and*

$$\int_a^b f = \int_a^c f + \int_c^b f$$

Proof. Let $f_1 = f|_{[a,c]}$ and $f_2 = f|_{[c,b]}$. Now $f \in \mathcal{R}[a,b] \implies$ there exists a partition $P = \{x_0, \ldots, x_n\}$ such that

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) < \epsilon$$

Let c be such that $x_{i-1} < c < x_i$. Let $P^* = P \cup \{c\}$. Then

$$\mathcal{U}(f,P^*) - \mathcal{L}(f,P^*) \leq \mathcal{U}(f,P) - \mathcal{L}(f,p) < \epsilon$$

Let $P_1 = \{x_0, \ldots, c\}$, $P_2 = \{c, \ldots, x_n\}$. With this, we have

$$\mathcal{U}(f,P^*) = \mathcal{U}(f_1,P_1) + \mathcal{U}(f_2,P_2)$$

and,

$$\mathcal{L}(f,P^*) = \mathcal{L}(f_1,P_1) + \mathcal{L}(f_2,P_2)$$

□

Proposition 5. *Let $f \in \mathcal{R}[a,b]$. Then*

1. *If $m \leq f(x) \leq M \ \forall \ x \in I = [a,b]$, then*

$$m(a-b) \leq \int_a^b f \leq M(a-b)$$

2. *If $g \in \mathcal{R}[a,b]$ such that $g \leq f \ \forall \ x \in I$, then*

$$\int_a^b g \leq \int_a^b f$$

3. *If $|f| \leq M$ then*

$$|\int_a^b f| \leq M(b-a)$$

Proof. Suppose $m \leq f \leq M$ on I. Let $P = \{a,b\}$. Now

$$\int_a^b f \geq \mathcal{L}(f,P) = (\inf_I f(x))(b-a) \geq m(b-a)$$

Similarly,

$$\int_a^b f \leq \mathcal{U}(f,P) \leq M(b-a)$$

Suppose now that $g \in \mathcal{R}[a,b]$ and $g \leq f$. Then just consider the function $h = f - g \geq 0$. By the previous property, we have

$$\int_a^b (f-g) \geq 0$$

and by linearity of the integral, we have (2). The proof of the third property is left as an exercise. □

Theorem 1. *Any continuous function on $[a,b]$ is Riemann Integrable.*

Proof. Let $f : [a,b] \to \mathbb{R}$ be continuous. Then f is bounded and f is uniformly continuous. Let $\epsilon > 0$ Then there exists some $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon$$

whenever $|x-y| < \delta$. Let $P = \{x_0 < \cdots < x_n\}$ be a partition of $[a,b]$ such that

$$\inf_{x \in I_k} f(x) = f(a_k)$$

and

$$\sup_{x \in I_k} f(x) = f(b_k)$$

for some $a_k, b_k \in I_k$ as f is continuous. Then

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) = |\sum_{k=1}^{n}(x_k - x_{k-1})(f(b_k) - f(a_k))| \leq \sum_{k=1}^{n}(x_k - x_{k-1})|f(b_k) - f(a_k)|$$

$$a_k, b_k \in [x_{k-1}, x_k] \implies |b_k - a_k| < \delta \implies |f(b_k) - f(a_k)| < \epsilon/(b-a)$$

and thus

$$\mathcal{U}(f,P) = \mathcal{L}(f,P) \leq \frac{\epsilon}{b-a}\sum(x_i - x_{i-1}) = \epsilon$$

and hence $f \in \mathcal{R}[a,b]$. □

Theorem 2. *Any monotone function on $[a,b]$ is Riemann Integrable.*

Proof. For simplicity, we assume f to be monotonically increasing.
Then $f(a) \leq f(x) \leq f(b)\ \forall\ x \in [a,b]$, and hence f is bounded.
Let $P = \{x_{-0} < \cdots < x_n\}$ be a partition such that

$$\sup_{x \in I_k} f(x) \leq f(x_k)$$

and

$$\inf_{x \in I_k} f(x) \geq f(x_{k-1})$$

$$\mathcal{U}(f,P) - \mathcal{L}(f,P) = \sum (x_k - x_{k-1})(\sup_{x\in I_k} f(x) - \inf_{x\in I_k} f(x))$$

$$\leq \sum (x_k - x_{k-1})(f(x_k) - f(x_{k-1})) < \frac{\epsilon}{f(b)-f(a)} \sum f(x_k) - f(x_{k-1}) < \epsilon$$

□

Homework: If $\int_a^b f = 0,\ f \geq 0$ and f is continuous on $[a,b]$. Then $f \equiv 0\ \forall\ x \in [a,b]$.

Proposition 6. *Let $f \in \mathcal{R}[a,b]$, $m \leq f(x) \leq M$ on $[a,b]$ and $\phi : [m,M] \to \mathbb{R}$ be continuous. Then $\phi \circ f \in \mathcal{R}[a,b]$.*

Proof. Let $\epsilon > 0$. Then there exists a $\delta > 0$ such that

$$|\phi(x) - \phi(y)| < \epsilon$$

whenever $|x-y| < \delta$. Now $f \in \mathcal{R}[a,b] \implies$ there exists a partition P such that $\mathcal{U}(f,P) - \mathcal{L}(f,P) < \epsilon$. Let Q be a refinement of P so that the length of each of the subintervals in Q is $\delta/4k$ where $k > 1$ and $k > \phi(x)$ for all $x \in [m,M]$. Now, $\mathcal{U}(f,Q) - \mathcal{L}(f,Q) < \epsilon$ and

$$\mathcal{U}(\phi \circ f, Q) - \mathcal{L}(\phi \circ f, Q) = \sum [\sup \phi(f(x)) - \inf \phi(f(x))](x_k - x_{k-1})$$

$$= \sum (M_k - m_k)(x_k - x_{k-1}), \text{ where } M_k = \sup_{x\in I_k} \phi(f(x)) \text{ and } m_k = \inf_{x\in I_k} \phi(f(x))$$

Let a_k, b_k be the supremum and the infimum defined over the k-th subintervals respectively.

□

Class 6, January 27 2026

Dirichilet's Test

Let $f, g : [a,\infty) \to \mathbb{R}$ for some $a \in \mathbb{R}$. Assume that g is monotonic and $\lim_{x\to\infty} g(x) = 0$. If f is continuous and if there is a positive M such that

$$\Big|\int_a^x f(t)\ dt\Big| \leq M$$

for all $x > a$ and $g(x)$ is differentiable. Then $\int_a^\infty f(x)g(x)\ dx$ converges.

Proof. We will use the fact that g is differentiable. We let

$$F(x) = \int_a^x f(t)dt$$

for all $x \geq a$. Then $|F(x)| \leq M$ for all $x \geq a$. Now

$$\int_b^c f(x)g(x)dx = F(c)g(c) - F(b)g(b) - \int_b^c F(x)g'(x)dx \text{ (using integration by parts)}$$

As F is bounded,

$$F(c)g(c), F(b)g(b) \to 0$$

as $b \to \infty$. Then

$$\left| \int_b^c F(x)g'(x)dx \right| \leq \int_b^c |F(x)| \cdot |g(x)|dx \leq M \int_b^c |g'(x)|dx$$

Since g is monotonic, g' does not change sign and thus we have

$$\int_b^c |g'(x)|dx = |\int_b^c g'(x)dx| = |g(c) - g(b)|$$

where the last term goes to 0 as $b \to \infty$. So the integral converges. □

Proposition 7. *Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Suppose $f \in \mathcal{R}[c, b]$ for all $c \in (a, b)$. Then $f \in \mathcal{R}[a, b]$ and*

$$\int_a^b f(x)\ dx = \lim_{c \to a} \int_c^b f(x)\ dx$$

Proof. Let $\epsilon > 0$. Let $M > |f(x)|$ for all $x \in [a, b]$. $f \in \mathcal{R}[a + \frac{\epsilon}{4M}, b]$. There is a parition P of this interval such that

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) < \frac{\epsilon}{2}$$

Let $\tilde{P} = P \cup \{a\}$. Then

$$\mathcal{U}(\tilde{P}, f) - \mathcal{L}(\tilde{P}, f)$$
$$= \big[\sup_{a \leq x \leq a + \frac{\epsilon}{4M}} f(x) - \inf_{a \leq x \leq a + \frac{\epsilon}{4M}} f(x) \big] + \mathcal{U}(P, f) - \mathcal{L}(P, f)$$

Now,

$$|\sup f(x)| \leq \sup |f(x)| \leq M \text{ for all } x$$

□

Theorem 3. *Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Suppose f is continuous on $[a, b]$ except possibly at finitely many points. Then $f \in \mathcal{R}[a, b]$.*

Proof. Suppose f is discontinuous on the set F and for simplicity assume that $F = \{a\}$. The idea is to break the entire interval at points of discontinuity. Then it is enough to show that f is Riemann integrable on an interval even if it is discontinuous at the endpoints. □

We know that if f is bounded on $[a, b]$ and $\int_c^b f(x)\ dx$ exists for any $c \in [a, b]$ then this integral converges to $\int_a^b f(x)\ dx$. But what if f is not bounded on the interval?
Let $f : (a, b] \to \mathbb{R}$ and $f \in \mathcal{R}[x, b]$ for every $x \in [a, b]$. We say that

$$\int_a^b f(x)\ dx$$

exists(or converges) if

$$\lim_{x \to a} \int_x^b f(x)\ dx$$

exists.
Example:

$$\int_\epsilon^b x^{-p}\ dx = \frac{b^{1-p} - \epsilon^{1-p}}{1-p}$$

for $p \neq 1$. Then the integral converges to $\frac{b^{1-p}}{1-p}$ as $\epsilon \to 0$.
Similarly, we can define $\int_a^b f(x)$

Class 7, February 3 2026

f_n is said to converge pointwise to f in E if for every $\epsilon > 0$ and and every $x \in E$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \epsilon$$

f_n is said to converge uniformly to f if the same N works for every $x \in E$.
We consider double sequences. Suppose $a_{m,n} = \frac{n}{n+m}$. Then $\lim_{n\to\infty} a_{m,n} = 1$ and $\lim_{m\to\infty} a_{m,n} = 0$. This shows that in general we cannot swaps double limits over a doubly indexed sequence. We cannot swap limits in pointwise convergence.
Take the example,

$$f_m(x) = \lim_{n\to\infty} (\cos m!\pi x)^{2n} = \begin{cases} 0 \text{ if } x \in \mathbb{Q} \\ 1 \text{ if } x \in \mathbb{Q},\ x = \frac{p}{q},\ m \geq q \end{cases}$$

Taking $m \to \infty$, we find

$$f(x) = \begin{cases} 0 \text{ if } x \notin \mathbb{Q} \\ 1 \text{ if } x \in \mathbb{Q} \end{cases}$$

Next, we consider the example

$$f_n(x) = n^2 x(1-x^2)^n\ ,\ x \in [0,1]$$

Then,

$$\lim_{n\to\infty} f_n(x) = 0$$

Uniform Convergence

Let E be a set and f_n be a sequence of $\mathbb{R}/\mathbb{C}$ valued functions on E. We say that f_n converges to uniformly to a function f on E if for each $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|f_n(x) - f(x)| < \epsilon$$

for all $x \in E$. In this case we write $f_n \xrightarrow{u} f$ on E. Clearly, for a finite set E, pointwise and uniform convergence are all the same since we are able to get a maximum N that works for all $x \in E$.

Proposition 8. *If $f_n \to f$ on E, then for each $x \in E$, $f_n \to f$ pointwise in E.*

We have a Cauchy Criterion here as well.

Theorem 4. *Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of functions on E. Then $\{f_n\}_{n\in\mathbb{N}}$ converges uniformly to a function f on E iff for each $\epsilon > 0$, there exists an $N \in \mathbb{N}$,*

$$|f_m(x) - f_n(x)| < \epsilon$$

for every $m, n \geq N$

Proof. ($\Longleftarrow$) Suppose $\{f_n\}_{n\in\mathbb{N}}$ converges uniformly to a function f on E. Let $\epsilon > 0$. Then by uniform convergence,

$$|f_n(x) - f(x)| < \epsilon$$

for every $x \in E$ and for all $n \geq N$ for some $N \in \mathbb{N}$ depending on our choice of ϵ. Then for any other $m \geq N$, we have by the triangle inequality,

$$|f_m(x) - f_n(x)| < 2\epsilon$$

Since ϵ was arbitrary, we have our result.
($\Longrightarrow$)Suppose to each $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that

$$|f_n(x) - f_m(x)| < \epsilon$$

for all $m, n \geq N$ and for all $x \in E$. Thus the sequence $\{f_n(x)\}_{n\in\mathbb{N}}$ is Cauchy for every x. Then $\{f_n(x)\}_{n\in\mathbb{N}}$ converges for every $x \in E$. Then we let $m \to \infty$, to get

$$-\epsilon \leq f_n(x) - f(x) \leq \epsilon$$

But since ϵ was arbitrary, we have uniform convergence. □

Theorem 5 (L^∞ Test)**.** *Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of functions on a set E and f be a function on E. Then $\sup_{x\in E} |f_n(x) - f(x)| \to 0$ iff $f_n \xrightarrow{u} f$.*

Proof. Let $f_n \xrightarrow{u} f$. So to each $e\epsilon$ there is an $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$ and $x \in E$.
($\Longleftrightarrow$) to each $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that

$$\sup_{x\in E} |f_n(x) - f(x)| \leq \epsilon$$

for all $x \in E$.
($\Longleftrightarrow$)

$$\sup_{x\in E} |f_n(x) - f(x)| \to 0 \text{ as } n \to \infty$$

□

Exercise: Show that if $f_n \xrightarrow{u} f$, then $|f_n| \xrightarrow{u} |f|$.
Exercise: If $f_n \xrightarrow{u} f$ then does $\sup_{x\in E} f_n \to \sup_{x\in E} f$?

Series

Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of functions on a set E. We say that $\sum_{n=1}^{\infty} f_n$ converges uniformly on E if the sequence of partial sums converges.

Theorem 6 (Weirstrass M-Test)**.** *Let $\{f_n\}_{n\in\mathbb{N}}$ be a sequence of functions and let $M_n = \sup_{x\in E} |f_n(x)|$. Then $\sum_{n=1}^{\infty} f_n$ converges uniformly if $\sum_{n=1}^{\infty} M_n < \infty$.*

Proof. Since $\sum_{n=1}^{\infty} M_n$ converges, for each ϵ there is an N such that $\sum_{k=n}^{m} M_k < \epsilon$ for all $m \geq n \geq N$. Then

$$|\sum_{k=n}^{m} f_k| \leq |\sum_{k=n}^{m} M_k| < \epsilon$$

Therefore, $\sum_{n=1}^{\infty} f_n(x)$ converges. □

The M-Test works only in one direction.
Example: $f_n(x) = \frac{x^{2n}}{n^2}$, $x \in [-1, 1]$. Now $M_n = \sup |f_n(x)| = \frac{1}{n^2}$. Then since the sum of this series converges, the original series converges as well.
Example:

$$f_n(x) = \begin{cases} 0 \text{ if } x \neq \frac{1}{n} \\ \frac{1}{n} \text{ if } x = \frac{1}{n} \end{cases}$$

But we find that the sum of supremums does not converge.

Theorem 7. *Let $f_n : [a, b] \to \mathbb{R}$ be a sequence of functions that converges uniformly to a function f on $[a, b]$. If $\lim_{t\to x} f_n(x) = A_n$ exists, for some $x \in [a, b]$ for all $n \in \mathbb{N}$ then*

$$\lim_{n\to\infty} A_n$$

exists and equals

$$\lim_{t\to x} f(t) = \lim_{t\to x} \lim_{n\to\infty} f_n(t)$$

Thus the limits can be exchanged.

Proof. First we show that $\{A_n\}_{n\in\mathbb{N}}$ converges. For this we show that it is Cauchy. Let $\epsilon > 0$, then there exists some N such that

$$|f_n(t) - f_m(t)| < \epsilon$$

for all $m, n \geq N$ and for all $t \in [a, b]$. Let $m, n \geq N$. Let $t \to x$ in the previous equation, then

$$\lim_{t\to x} |f_m(t) - f_n(t)| \geq \epsilon$$

and since the absolute value is continuous, we have

$$|A_m - A_n| \leq \epsilon$$

. This implies that the sequence is Cauchy. Then $\{A_n\}_{n\in\mathbb{N}}$ converges. Let $A = \lim A_n$. Then there is an N such that $|f_n(t) - f(t)| < \frac{\epsilon}{3}$ for all $n \geq N$,

$t \in [a,b]$ and $|A_n - A| < \frac{\epsilon}{3}$ for all $n \geq N$. Now $\lim f_n = A_n \implies$ there is a $\delta > 0$ such that $|f_N(t) - A_N|$ for all $t \in (x-\delta, x+\delta) \cap [a,b] - \{x\}$. Finally,

$$|f(t) - A| \leq |f(t) - f_N(t)| + |f_N(t) - A_N| + |A_N - A| < \epsilon$$

for all $t \in (x-\delta, x+\delta) \cap [a,b] - \{x\}$. From this, our result follows easily. □

Corollary 1. *Uniform limit of continuous functions is continuous.*

Proof. Let $\{f_n\}$ be a sequence of continuous functions on an interval I such that $f_n \xrightarrow{u} f$. Let $x \in I$, for all n, we have

$$\lim_{t \to x} f_n(t) - f_n(x) \implies \lim_{t \to x} f(t) = \lim_{n \to \infty} f_n(x) = f(x) \implies f \text{ is continuous at } x$$

□

Class 8, February 6 2026

Corollary 2. *Let $\{f_n\}$ be a sequence of functions in an interval I. Suppose $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly and each of the $f_i(x)$ are continuous. Then their sum is also continuous.*

Proof. Let s_n be the n-th partial sum. Then $s_n \to \sum_{n=1}^{\infty} f_n$. Now, the uniform limit of continuous function is continuous. Thus our result holds. □

Theorem 8. *There exists a continuous function that is nowhere differentiable.*

Proof. Let $f(x) = |x|$ defined on $[-1,1]$. Then f is not differentiable just at one point. Now define $\phi : \mathbb{R} \to \mathbb{R}$ as

$$\phi(x) = \begin{cases} |x|, \ |x| \leq 1 \\ |x+2k|, \ |x+2k| \leq 1 \text{ where } k \text{ is an integer} \end{cases}$$

ϕ is the extension of f with $\phi(x+2) = \phi(x)$. Therefore, ϕ is continuous and one can verify that $|\phi(x) - \phi(y)| \leq |x-y|$. Let $g(x) = \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \phi(4^n x)$ Now,

$$|\left(\frac{3}{4}\right)^n \phi(4^n x)| \leq \left(\frac{3}{4}\right)^n$$

which leads to us conclude that g converges uniformly and thus is continuous. Now let $x \in \mathbb{R}$. Fix $n \in \mathbb{N}$. Let $\delta_m = \pm\frac{1}{2}4^m$ where sign is chosen so that there are no integers between $4^m x$ and $4^m(x+\delta_m)$. Let $\gamma_m = \frac{\phi(4^m(x+\delta_m)) - \phi(4^m x)}{\delta_m}$ For $n > m$, $4^n \delta_m$ is an even integer and $\phi(4^n(x+\delta_m)) = \phi(4^n x)$. Therefore, $\gamma_n = 0$. Thus we have

$$\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \frac{\phi(4^n(x+\delta_m)) - \phi(4^n x)}{\delta_m}$$

One can conclude now that g is not differentiable anywhere. □

Corollary 3. *Let $\{f_n\}$ be a sequence of Riemann Integrable functions such that their sum converges uniformly on $[a, b]$. Then $\sum_n f_n \in \mathcal{R}[a, b]$ and*

$$\int_a^b \sum_n f_n = \sum_n \int_a^b f_n$$

Theorem 9. *Let $\{f_n\}$ be a sequence such that*

1. *Each f_n is continuous on $[a, b]$.*
2. *f_n converges to f pointwise for all $x \in [a, b]$ for some continuous function f.*
3. *f_n is a decreasing sequence.*

Then

$$\int_a^b f_n = \int_a^b f$$

This is analogous to MCT. In fact, $f_n \xrightarrow{u} f$

Proof. Assume first that f is identically 0 on $[a, b]$. Let $g_n = f_n - f$. Then $g_n(x) \geq g_{n+1}(x)\ \forall\ x \in [a, b]$ and $g_n(x) \to 0$. Let $\epsilon > 0$ and let$k_n = \{x : g_n(x) > \epsilon\}$. Then we claim that some k_n are empty. If this were false, assume that all k_n were non-empty. Suppose $x_n \to x$. Then $x_n \in k_n\ \forall\ n$. Since $x_n \in k_n$, $g_n(x_n) > \epsilon$. Then letting $n \to \infty$ we find that $x \in k_n$ for all n. That means that $g_m(x) > \epsilon$ for all m for some x. But this is a contradiction to the fact that g_n converges to 0. Therefore, k_n is empty for some n and since this is a decreasing sequence, we conclude that k_m is empty for all $m \geq n$ and all $x \in [a, b]$. Therefore, g_n converges to 0 uniformly. $\square$

Example: Let $f_n(x) = \frac{1}{nx+1}$. Note that f_n is decreasing but does not converge unformly. This is fails since this is considered in an open interval.